

# Improving Efficiency, Reliability and Life-time Cost of Data Centers Using DC Technology

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**Abstract**— With the expansion of today's digital era, the use of electronic equipment is increasing exponentially at home, office, and plants. The common denominator of these electronic equipment is that they all use Direct current (DC) power internally by converting Alternating Current (AC) power to DC power using local converters. All these converters produce losses, heat, and are prone to failures which in turn impact the equipment efficiency, life span, and reliability. In addition, these converters increase the cost of equipment they are powering and their required footprint. The issue of reliability and cost reduction is more stressing for data centers where servers, Uninterruptable Power Supply (UPS) systems, and IT equipment constituting the majority of the load can be powered directly through a DC grid. This paper presents a case study that compares AC and DC technologies for powering a typical data center and outlines the main advantages and limitations of each technology.

This study will present a comparative analysis of a DC power system against a typical AC distribution for critical datacenter loads. It also presents a real case study of existing AC powered data center and compare the benefits of powering the center via DC instead of AC. Analytical reliability, safety and efficiency models are developed, and quantitative results are obtained based on a set of real data. The study indicates that the DC system would be more reliable, efficient and will achieve less Capex and Opex values than the AC.

**Keywords**—AC Architectures, Reliability, Efficiency and Total Cost of Ownership

## I. INTRODUCTION

The current industrial revolution and the rapid increase of the digital economy is driving fast development in data center industry as one of the cornerstones of the world economy. This growth in data centers is putting some stresses on several aspects of this industry. These stresses vary from meeting the industry energy demand, reducing environmental impact, securing lands and services, raising the need for cybersecurity and others.

The above stressing factors create more concerns on the way that current data centers are built with. Hence, more emphasis

for better efficiency, more balanced between reliability and cost effectiveness and more intelligence in power distribution.

This paper will examine the current data centers AC power architecture and compare it with DC power architecture as a viable alternative to analyze the merit of such architecture deployment in facing the current industry growth and overcome its challenges. The paper will compare both architectures against reliability, efficiency, power quality, safety and footprint aspects. After that, a cost analysis is performed to evaluate which architecture has lower Total Cost of Ownership (TCO) to achieve the right balance between cost effectiveness and all other aspects.

## II. DRIVERS OF DC POWER IN DATA CENTERS

The current dramatic increase in data center industry and the high demand of energy consumption created both financial and environmental challenges. Also, the integration with renewable sources together with the need of maximizing data centers power efficiency, to deliver more densities and meet the computing demand, put more emphasis on the development of a modern data centers from design and operation aspects.

Although all data centers IT loads such as CPUs and drives consume DC power, such facilities are powered by AC due to its simplicity of converting and transmitting power to the IT loads. Modern power electronics devices have the same AC power capability with higher efficiency and compatibility with DC generated power through renewable sources.

DC power becomes viable option in powering data center as it offers many features compare to the AC such as suffers lower electrical losses (inductive & capacitive currents), allows 29% more effective voltage than AC cable of equal rating, no reactive power losses, no need for reactive power compensation, reduced number of conductors, reduced RI2 losses, reduced transmission corridors, no synchronization

components and operations, fewer components, lower conversion stages and less footprint required.

### III. AC & DC DISTRIBUTION ARCHITECTURES

There are many AC and DC architectures in data center industry. In this paper, the widely used AC architecture which is illustrated in Fig. 1 will be analyzed and compared to other DC architectures in terms of reliability, power quality, energy efficiency, safety and footprint. In this paper's case study, this AC architecture will be compared to the DC one showed in Fig.2. This is to compare in-kind architectures in order to properly identify which architecture has an edge over the other.

The below AC architecture in Fig.1 is a widely used one which has an AC UPS at 480V (or other standard distribution voltage level) where the power will be converted twice to charge the battery system via DC power and supply the AC Power Distribution Units (PDU). After which the AC power will be step down to the rack level voltage. The server Power Supply Unit (PSU) will again lower the input voltage and converted to DC to the IT components.

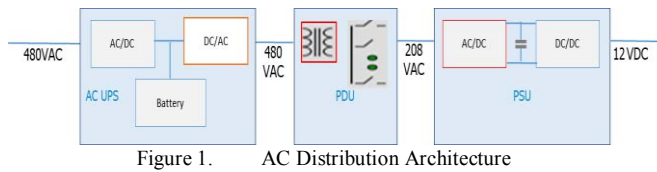


Figure 1. AC Distribution Architecture

Similarly, the below DC architecture in Fig.2 performs exactly what mentioned above but with lower conversion and transforming stages and will require the server to accept DC at its input.

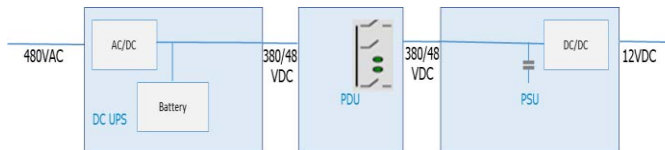


Figure 2. DC Distribution Architecture

DC power can be delivered to the IT load via several architectures. One of which is the High Voltage DC (HVDC) where DC power will be distributed at higher voltage level. Another architecture is to run AC all the way toward the white space and make the conversion either at the white space level or at the rack level. Other architectures such as LVDC 48V or 12V are also widely used in telecom industry. Fig.3 has some of the above stated DC architectures.

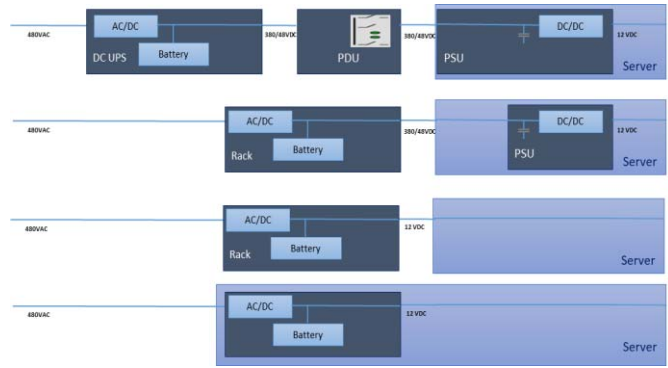


Figure 3. Different Types of DC Distribution Architecture

### IV. CLOSER LOOK INTO AC & DC SYSTEM PROS AND CONS

In the below sections, AC & DC architectures will be evaluated and compared from different aspects which are reliability, power quality, energy efficiency, safety and footprint.

#### A. Reliability

The reliability of a system is its ability to perform required function. The higher the number of series components in a distribution path, the less the reliability in other words an existence of single point of failure in the system will reduce the system reliability. Also, the fewer the components needed to perform the required function, the higher the reliability. These are some of the general rules of IEEE 493 [1].

The basic Reliability method is the reliability block diagram (RBD) which looks after the series and parallel components in the system such as UPS, battery, generator and others. The calculated reliability through RBD is 37.2% for AC architecture compare to 81.5% for DC architecture [8]. This is to get a basic idea about system reliability. However, in data center -where there are large number of components, different sequence of operation, existence of standby equipment and others- a more comprehensive analysis shall be performed such as the Monte Carlo simulation. The result of different studied AC and DC architectures showed that the failure rate per year in DC architecture is ranged from 20-40% less than the AC architecture [2]. Hence, this simulation results proved that the DC architectures -in general- have higher reliability level than AC. It is worth mentioning that the reliability level gap between DC & AC will be disappear at higher redundancy level of AC architectures.

#### B. Power Quality

The power quality level is of a paramount impact on data center operations. So, power quality disturbances and system performance shall always be studied and mitigated. There are some common power quality disturbances that occur in AC and DC which will not be discussed here as their impact and mitigation are similar. So, the focus will be on what power quality merits and issues that DC architecture has.

The Power quality disturbance in DC architecture is significantly less than AC as there are less harmonics, no unbalance loads issue, no power factor correction circuits required and no need for synchronization circuits [3]. In contrast, continuous change and switching in IT loads will create ripples in the DC bus. This can be easily mitigated by the battery system via supplying the needed current to stabilize the bus voltage. Furthermore, voltage sags and DC bus voltage can be managed by the rectifier.

### C. Energy Efficiency

Efficiency and energy delivery to the IT loads are crucial aspects to all data centers owners. AC equipment is currently achieving the highest efficiency level and can run on different operational mode to become even more efficient. Nevertheless, the lower conversion stages and components in DC architectures will always be a positive aspect of DC architectures.

Several recent industry studies that compare AC to DC efficiency concluded that DC is more efficient than AC. Some indicated that DC is 10% more efficient than AC [3]. Other concluded that DC achieves 17% Energy Reduction compared to AC and power loss is reduced by 36% in DC compared to AC [7]. On the other hand, this paper indicates different figure, too, as detailed in the case study section. The take away here is that DC is always more efficient than AC and the data center architecture and eco-system what basically determine by how much.

It's worth mentioning that efficiency has a cascaded impact since it is all about the whole ecosystem not only the individual equipment. For instance, DC suffers less conductor losses than AC. This results in 1-2% higher efficiency in DC architecture [5]. Also, the higher the efficiency, the lower heat dissipated and the resulted cooling power. Some studies concluded that about 11% efficiency gain will result in a cost saving of about \$740,000 in three years period [6].

### D. Safety

The topic of DC voltage is not new to the industry as it is already being used in many industries such as microgrids and smart cities, HVDC system that interconnect different grids together, telecom, railways [4], renewable, etc. Thus, experience has been gained in dealing with DC systems. The concern of safely interrupting DC current is presently vanished by the development of power electronic devices that has fast interrupting capability, DC breakers technology and Residual Circuits Breakers (RCB). However, safety concerns are of an equal criticality for AC & DC that is why NFPA and other industry standards classify the arcing levels and their mitigation such as remote racking, switching and other.

In regard to personnel protection, it is important to know that the impact of electric shock depends on current magnitude, duration, current path and type of voltage AC or DC. Thus,

the magnitude of safe operating current of DC is always higher than AC which makes DC safer.

### E. Footprint

Since DC architectures have fewer components than AC, a smaller physical footprint is required. This will result not only in delivering more power capacity as a result of reduced losses but also will reduce the cost of data center building and land which represent an advantage of lower TCO in DC data centers.

## V. CASE STUDY. TOTAL COST OF OWNERSHIP (TCO) COMPARISON FOR DC AND AC ARCHITECTURES

The study presents an energy efficiency analysis which has been conducted for an existing AC powered data center and compared against a similar-in-kind modeled DC powered data center via utilizing Etap simulation software.

The cost analysis in this paper is focused on the total cost of ownership which covers both the Operational Expenses (Opex) and the Capital Expenses of the equipment (Capex). The Opex analysis has been simplified to cover only the power losses as a result of the conversion stages in both AC and DC systems. Although cables' losses and number of conductors in DC system are less compared to AC, these losses have been excluded from the study and considered alike due to the fact that different DC distribution architectures will result in different analysis results. Also, and for the same reason maintenance costs were exempted. On the other hand, the capex analysis covers the conversion equipment costs and disregards the building and land cost while of course the footprint in DC is smaller than the AC. The objective of this analysis is to set a baseline mature AC architecture and compare it with similar DC architecture via eliminating the AC converters. Thus, the selected DC architecture was the exact AC one without some conversions.

The study analyzes the above stated AC distribution architecture in Fig.1 as a baseline and compares it to the DC distribution architecture in Fig.2. The single line diagram of AC architecture is shown in Fig.4. It represents a tier VI data center with a redundancy level of 2(N+1). Hence, the Site electrical system consists of 13.8 kV double ended switchgear with double tie breakers, two redundant UPS sub-systems, each of which contains 5 UPSS rated 0.48 kV, 750 KVA. Each UPS sub-system deliver the load to 12 PDUs (24 PDUs in total). The studied data center has three UPS systems with total UPS number of 30 units.

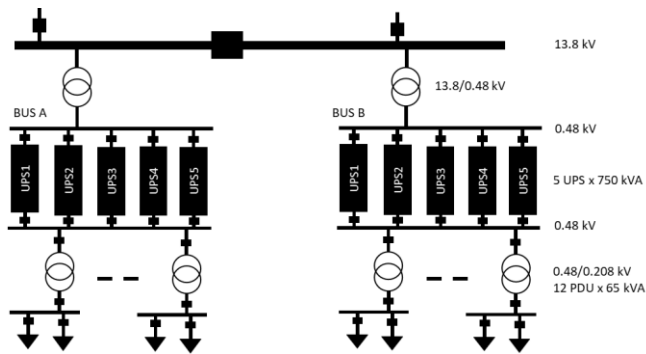


Figure 4. AC Architecture Single Line Diagram

The above redundancy level will result in a single UPS maximum loading of 25% which equivalents to energy efficiency of 89% as per the equipment manufacturer efficiency curve which is illustrated in Fig. 5 and validated by the site data. Normally, the UPS will be loaded to less than 15%. However, the study has taken the efficiency of the UPS as 91% considering efficient utilization of the UPS in its eco mode and not the status quo of 2 (N+1) configuration.

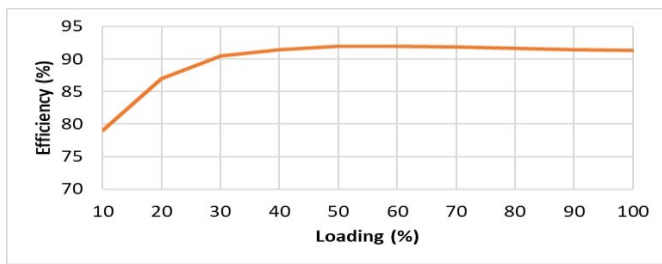


Figure 5. UPS Efficiency Curve

Etap software has been utilized for the load flow analysis where the data center has been modeled in the software. The total power consumption per single UPS system is 2.656 MW distributed between the sub-systems as illustrated in Table I.

TABLE I. SAMPLE OF LOAD FLOW ON THE SYSTEM INCOMERS

| Bus ID    | MW Loading | Mvar Loading | P+jQ         | S (MVA) |
|-----------|------------|--------------|--------------|---------|
| INCOMER_A | 1.327      | 0.683        | 1.327+0.683i | 1.492   |
| INCOMER_B | 1.329      | 0.683        | 1.329+0.683i | 1.494   |

The study parameters are described in the below Table II. It's worth mentioning that the \$/KWH cost as based on average tariff in the region, cooling ratio represents the cooling power requirements to remove the heat, UPS and PDU Transformer costs are comprehensive estimated cost based on combined suppliers' and data centers owners' inputs.

TABLE II. SAMPLE OF LOAD FLOW ON THE SYSTEM INCOMERS

| #  | Item                                       | Value      | Unit           |
|----|--------------------------------------------|------------|----------------|
| 1  | Operation Per Day                          | 24.00      | hours          |
| 2  | Operation Per Year                         | 365.00     | days           |
| 3  | Lifetime of The System                     | 25.00      | years          |
| 4  | Input Power                                | 2,656.00   | kW             |
| 5  | Cost of KWH (Average Tariffs)              | 0.13       | \$/kWh         |
| 6  | <b>VI. COOLING RATIO</b>                   | 60         | %              |
| 7  | 750KVA UPS Cost (Estimated)                | 600,000.00 | USD            |
| 8  | 300 kVA PDU's Transformer Cost (Estimated) | 7,500.00   | USD            |
| 9  | UPS Footprint                              | 15.21      | m <sup>2</sup> |
| 10 | PDU Transformer Footprint                  | 7.29       | m <sup>2</sup> |

The opex which focuses only on the losses costs has been illustrated in the below Fig. 6 and concluded that the DC architecture is 95.22% efficient compare to 88.92 % for AC architecture. The losses are driven by the conversion stages in AC such as the inverters and PDUs transformers which are not parts of the DC architecture. Also, this lower losses in the DC architecture leads to a reduced heat dissipation and in deed cooling requirements.

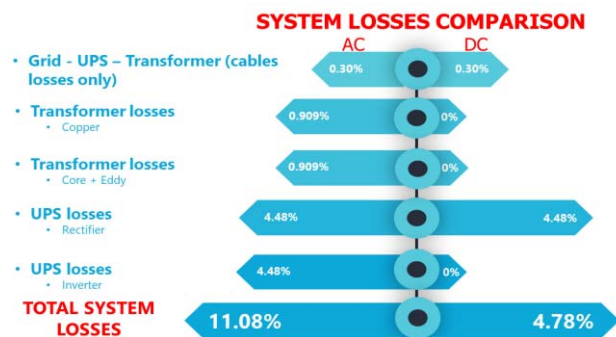


Figure 6. AC & DC Systems Losses Comparison

By converting these losses figures to \$ value, the total cost of losses of AC and DC architectures are listed in Fig.7. Thus, total cost saving in DC due to better efficiency and less conversions is \$7,586,421. For comprehensive and specific DC architecture, the saving will be much higher due to lower cabling losses and maintenance costs.

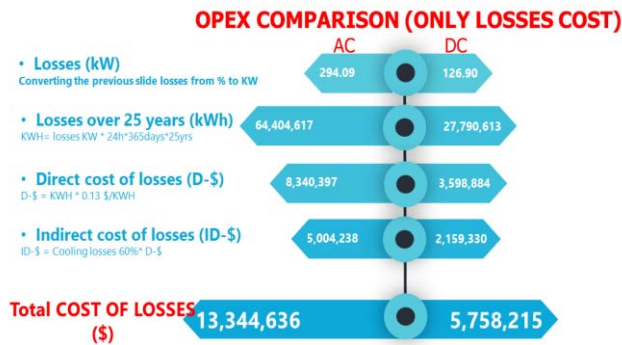


Figure 7. AC & DC Systems Opex Comparison

The capex cost analysis is illustrated in the below Fig. 8. The costs are associated with the UPSs and PDUs transformers for the AC and the Rectifiers in DC. Thus, the total cost saving in DC due to less capex is \$4,134,000. The capex in DC is expected to be lower due to the elimination of the inverters and PDUs' Transformers. In other DC configurations the saving will be even more. The more elimination of components the higher the saving.

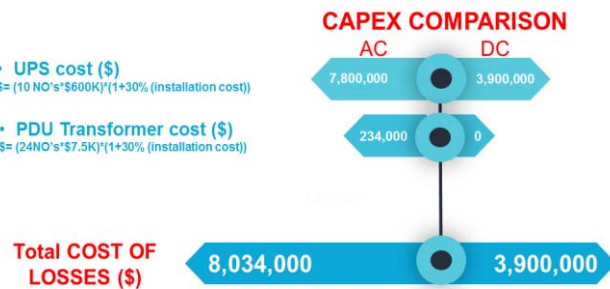


Figure 8. AC & DC Systems Capex Comparison

It's clearly observable that the Total Cost of Ownership (TCO) in DC is less than the AC and the cost saving for the life time of a single 2 (N+1) UPS system is \$11,725,291.

The cost analysis results show that DC architecture has 55% lower TCO compared to AC architecture and this value could be higher with other hybrid or Open Compute Project (OCP) architectures [8].

Another major cost saving factor is the footprint reduction in DC architecture due to reduced conversion stages. For this paper study and since the AC UPS footprint is 15.2m<sup>2</sup> compare to DC UPS with 7.6m<sup>2</sup> and the elimination of the PDU transformer with footprint of 7.29m<sup>2</sup>. The total saving in DC UPS footprint is about 76% of AC UPS plus the PDU.

## VII. DC ADAPTATION CHALLENGES

With the current direction toward DC power in many industries, data center industry is still facing several challenges to make this transformation despite all of the clear advantages of DC architectures over AC.

One of the challenges is the high maturity level and significant gained experience of data centers personnel in dealing with AC architectures. This helps achieve high availability and reliability levels in which all data center owners strive to sustain. All directives toward achieving higher efficiency levels were focused on data center components where several advancements to the AC electrical and mechanical equipment accomplished and indeed resulted in higher efficiencies of the whole ecosystem. However, the calls for reducing energy consumption and the limitation in energy delivery are more stressing these days due to the previously stated drivers toward DC power.

On the other hand, there are several barriers that delay DC architectures deployments such as lack of such DC inclusion in data center standards, Lack of DC ecosystems classifications and redundancy level and the majority of IT load accepts only AC power and convert it to DC.

Challenges are always there and a collaboration between data center industry owners, manufacturers, designers, consultants and governors is a must to overcome all challenges toward such transformation and help data center industry meeting the future demands.

## VIII. CONCLUSION

The DC architectures in data center is emerging with several demonstration sites and the deployments in other industry. The represented benefits and cost saving of DC over AC in this paper are key areas of focus and such analysis will continue to rationalize the transformation. As stated in this paper, many studies have listed some benefits and figures of DC data center from all aspects such as safety, reliability, efficiency and other.

The objective of this paper was to emphasize on the pros and cons of each architecture and most importantly to quantify the merits and cost advantage of DC architecture over AC. Finally, a collaboration between all data center industry chain will smooth such transformation and help boost the digital economy and the industry revolution.

## IX. REFERENCES

- [1] IEEE 493-2007 - IEEE Recommended Practice for the Design of Reliable Industrial and Commercial Power Systems
- [2] Bijen Raj Shrestha, Timothy M. Hansen and Reinaldo Tonkoski, "Reliability Analysis of 380V DC Distribution in Data Centers".
- [3] Satish Rajagopalan, Brian Fortenbery and Dennis Symanski, "Power Quality Disturbances Within DC Data Centers".
- [4] Tilo Pueschel, "DC-Powered Office Buildings and Data Centres".
- [5] Dustin J Becker and B.J. Sonnenberg, "DC Microgrids in Buildings and Data Centers".
- [6] Subrata Mondal and Earl Keisling, "Efficient Data Center design using Novel Modular DC UPS, Server Power supply with DC voltage and Modular CDU cooling".
- [7] Keiichi Hirose, "DC powered data center with 200 kW PV panels"
- [8] Alexander Barthelme, Xiwen Xu and Tiefert Zhao, "A Hybrid AC and DC Distribution architecture in Data Center"